

PROJECT REPORT

Automated Network Request Management System

Built on the ServiceNow Platform

Date	3 October 2026
Team ID	
Project Name	Automated Network Request Management System
Platform	ServiceNow (Service Portal, Service Catalog, Flow Designer)

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1. INTRODUCTION

1.1 Project Overview

The Automated Network Request Management System is an end-to-end Service Catalog solution built on ServiceNow. Employees raise requests for network access, Wi-Fi configuration, VPN provisioning or static IP allocation through a single Network Request catalog item in the Service Portal (/sp). Catalog UI Policies validate the form as the user types, a Flow Designer flow creates an auto-numbered record (DAT... prefix) in the custom table u_database_tables, the request is routed to the manager for approval, and automated email notifications are generated through sys_email. Every request is tracked with REQ... and RITM... numbers, so employees can see its progress without contacting IT.

1.2 Purpose

The purpose of the project is to replace unstructured email and paper-based network requests with a standard, validated and traceable process. The goals are:

- capture complete and correct network details at submission time;
- remove manual re-entry and manual record creation by IT;
- route approvals instantly to managers and keep a record of every decision;
- give employees transparent status tracking and automated email confirmations.

2. IDEATION PHASE

2.1 Problem Statement

Manual network request processing leads to delayed access provisioning, lack of standardized inputs, high rates of form errors, and poor visibility for tracking approvals and ticket progress across enterprise teams. Employees need a fast, error-free way to request connectivity and to know where their request stands; approvers and IT analysts need complete requests and clear notifications.

2.2 Empathy Map Canvas

The empathy map below captures what the primary user, an enterprise employee who needs network connectivity, says, thinks, does and feels before the solution is in place.

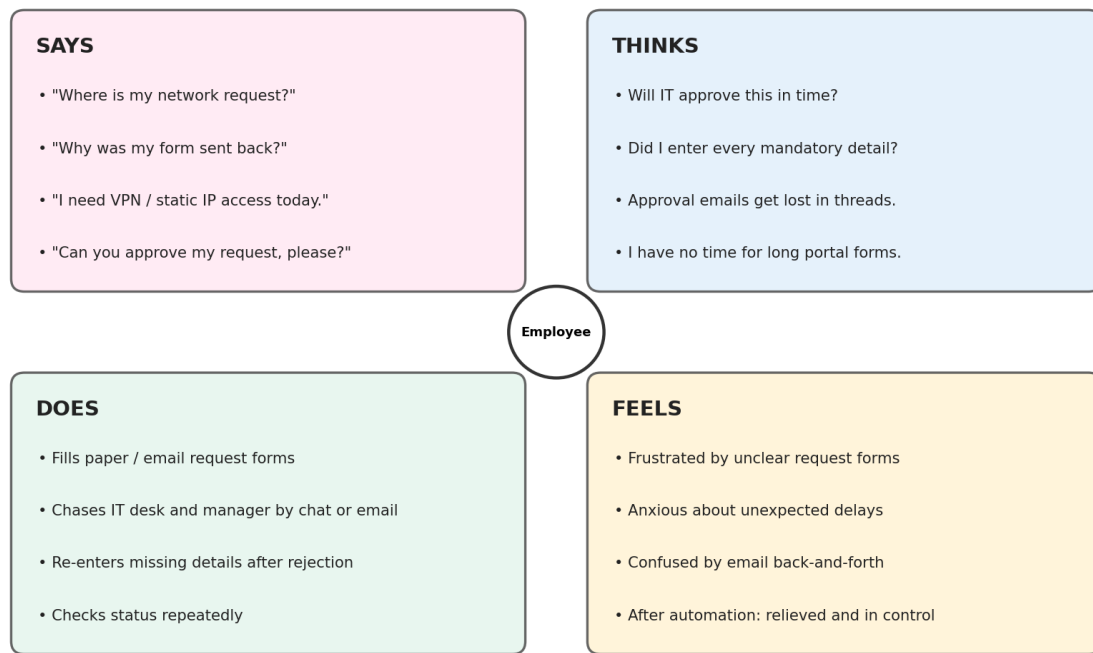


Figure 2.1: Empathy map of the enterprise employee

2.3 Brainstorming

The team brainstormed ideas around the three main problems: incomplete requests, slow manual approvals and no visibility. Ideas were then grouped and prioritised; the ones marked 'Selected' form the current release.

Idea	Theme	Decision
Single Network Request catalog item with dynamic variables	Incomplete requests	Selected
Catalog UI Policy to show/hide the ID field and toggle mandatory fields	Incomplete requests	Selected
Auto-numbered records (DAT...) in a custom table u_database_tables	No visibility	Selected
Flow Designer flow: get variables, create record, ask approval, update record	Slow approvals	Selected
Automated, dynamic HTML email on creation and approval (sys_email)	No visibility	Selected
Request tracking through REQ / RITM numbers and Request Summary page	No visibility	Selected
SLA timers and escalation for pending approvals	Slow approvals	Future scope
Virtual Agent / chatbot for status questions	No visibility	Future scope
Dashboard of request volumes and approval times	No visibility	Future scope
Automatic provisioning through integration with network devices	Slow approvals	Future scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey of an employee who needs a network connection, from the trigger to closure, with the automated solution in place.

Stage	Employee action	Touchpoint	Emotion	Pain point removed / opportunity
1. Need arises	Onboarding, location change, remote work or a failed connection	Line manager, IT notice	Worried	Trigger identified: a request is needed
2. Find the request	Searches 'network request' in the Service Portal and opens the item	Service Portal /sp	Neutral	One standard entry point instead of email
3. Fill the form	Enters requester information, mobile number, connection type, amount, payment mode, address	Catalog form + UI Policy	Confident	Dynamic validation prevents missing details
4. Order	Clicks Order Now and sees the Order Confirmation	Service Portal	Relieved	REQ / RITM numbers issued instantly
5. Approval	Manager receives the approval request; record is updated on decision	Flow Designer, approval	Waiting	No manual email chasing
6. Notification and tracking	Receives 'created' and 'approved' emails; checks Request Summary	Email (sys_email), Request Summary	Satisfied	Full transparency, no status calls

3.2 Solution Requirement

Functional requirements

ID	Functional requirement	Source / user story
FR-1	Employees can open the Network Request catalog item and fill the request form	USN-1, USN-2
FR-2	The form validates input dynamically (show/hide ID field, mandatory fields by connection type)	USN-7 to USN-10
FR-3	Order Now creates a request (REQ) and requested item (RITM) with a Request Summary	USN-11
FR-4	A flow creates an auto-numbered record (DAT...) in u_database_tables from the catalog variables	USN-12, USN-13
FR-5	The request is sent to the manager for approval and the record is updated on approve / reject	USN-14, USN-15
FR-6	Automated email notifications are sent when a request is created and when it is approved	USN-17, USN-18
FR-7	Employees and IT can track request status and view records in the Database Tables list	USN-19, USN-20

Non-functional requirements

ID	Requirement	Description
NFR-1	Usability	A single, short form that works in a standard desktop web browser with no extra plug-ins
NFR-2	Security	Role-based access and manager approval as required by organisational compliance

ID	Requirement	Description
NFR-3	Reliability	Every request produces a record and notifications; verified for all 3 test requests
NFR-4	Performance	Notification generated within about 1 second of record creation (test result: under the 3-second target)
NFR-5	Scalability	Cloud (SaaS) platform able to process many concurrent requests across departments
NFR-6	Availability	Hosted by the ServiceNow cloud platform, accessible through the Service Portal

3.3 Data Flow Diagram

Level 1 data flow: the employee submits the request; the order is processed, the flow stores the record in u_database_tables, the manager approves, and notifications are logged in sys_email.

Data Flow Diagram (Level 1) – Automated Network Request Management System

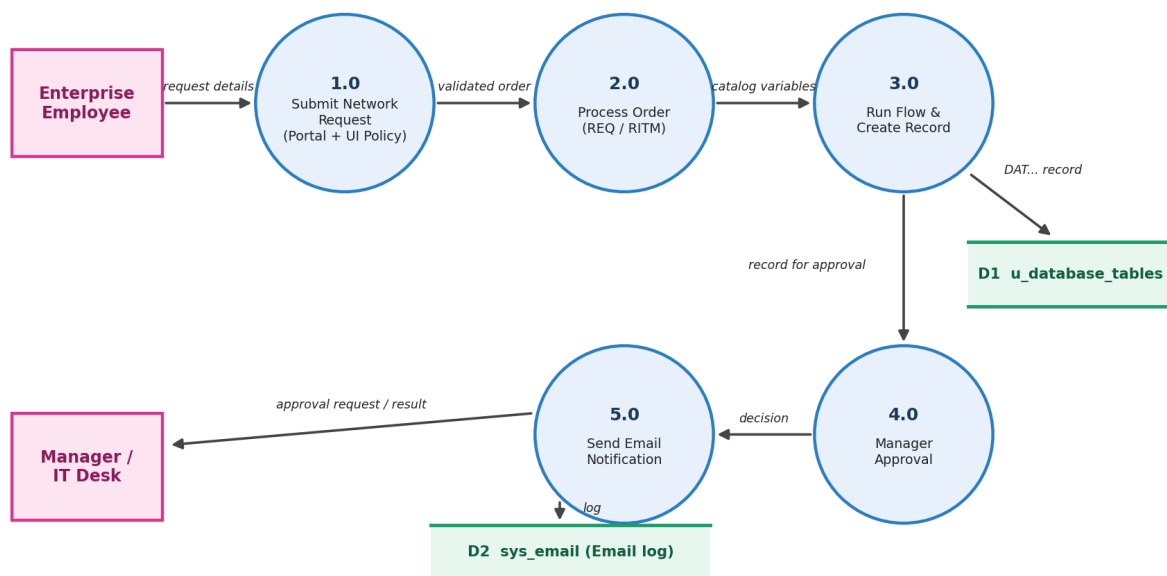


Figure 3.1: Data flow diagram (Level 1)

3.4 Technology Stack

Layer	Technology / component
Platform	ServiceNow (cloud SaaS) – developer instance
User interface	Service Portal (/sp), Service Catalog item 'Network Request', Variable Set 'Requester Information'
Validation	Catalog UI Policy ('Hide/Show of ID field'), Catalog Client Scripts framework
Automation	Flow Designer / Workflow Studio flow 'Network Request' (6 actions with an approval condition)
Database	Custom table u_database_tables (Database Tables) with DAT auto-number
Notification	Email notifications through sys_email (SMTP), dynamic HTML body
Client	Desktop web browser (Chrome)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit canvas confirms that the solution fits the customer's problem. Summary of the ten canvas sections:

Section	Summary
Customer Segments	Primary: enterprise employees. Secondary: IT service desk analysts, network engineers, department managers
Jobs-to-be-done / Problems	Submit network requests without errors; track progress and get instant approval updates; capture mandatory network information accurately
Triggers	Onboarding, location change, remote access need, access restrictions or hardware failures, project milestones
Emotions (before / after)	Before: frustrated, anxious, confused. After: relieved, confident and in control
Available solutions	Manual email / paper forms and basic ticketing: simple to start but error-prone, with delayed approvals
Customer constraints	Compliance and role-based security, little time, no third-party plug-ins
Behaviour	Users chase IT for status; approvers miss notifications in email threads
Channels	Online: Service Portal, email, browser. Offline: line manager and IT desk conversations
Root cause	No client-side validation; disconnected workflow needing manual IT intervention
Our solution	Network Request catalog item with UI Policies plus Flow Designer automation and email notifications

4.2 Proposed Solution

Parameter	Description
Problem statement	Manual processing causes delays, errors and poor visibility of approvals
Idea / solution	ServiceNow Service Catalog solution with dynamic forms, Flow Designer automation, DAT... records, manager approval and automated email
Novelty	Client-side UI policies integrated with backend flow orchestration, real-time auto-numbering and dynamic HTML emails with zero manual IT intervention
Customer satisfaction	Fulfilment times reduced from days to hours, no manual re-entry, full status transparency
Business model	Internal service-management ROI: lower IT support cost and administrative overhead
Scalability	Cloud SaaS architecture able to handle thousands of concurrent requests

4.3 Solution Architecture

Automated Network Request Management System - Solution Architecture

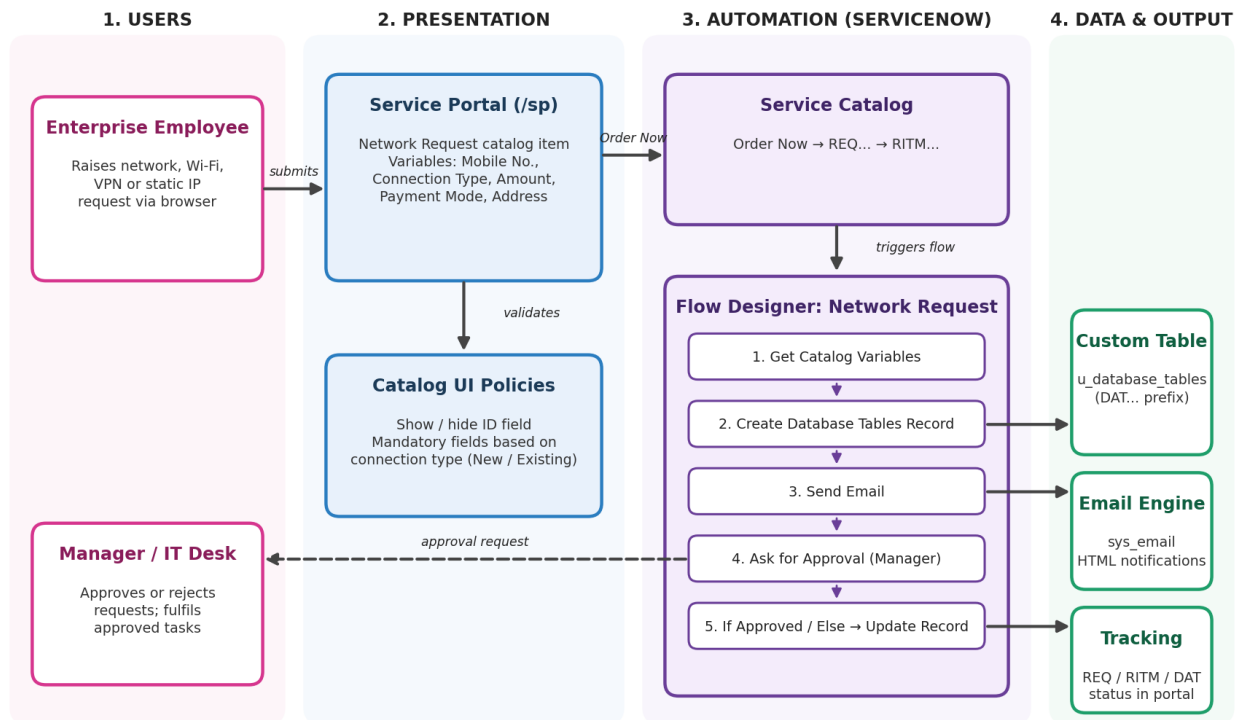


Figure 4.1: Solution architecture – users, presentation, automation and data layers

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was planned in four sprints of six days each (80 story points in total, 20 per sprint) using 22 user stories (USN-1 to USN-22).

Sprint	Epic(s)	User stories	Points
Sprint-1	Network Request Catalog Item; Database Table (u_database_tables)	USN-1 to USN-6	20
Sprint-2	Dynamic Form Validation; Request Submission	USN-7 to USN-11	20
Sprint-3	Workflow Automation (Flow Designer)	USN-12 to USN-16	20
Sprint-4	Email Notifications; Request Tracking; Testing and Deployment	USN-17 to USN-22	20

Sprint	Total story points	Duration	Start date	End date (planned)	Points completed	Release date
Sprint-1	20	6 days	07 Sep 2026	12 Sep 2026	20	12 Sep 2026
Sprint-2	20	6 days	14 Sep 2026	19 Sep 2026	20	19 Sep 2026
Sprint-3	20	6 days	21 Sep 2026	26 Sep 2026	20	26 Sep 2026
Sprint-4	20	6 days	28 Sep 2026	03 Oct 2026	20	03 Oct 2026

Velocity: team velocity is 20 story points per sprint and the sprint duration is 6 days, so the average velocity is $AV = 20 / 6 \approx 3.33$ story points per day.

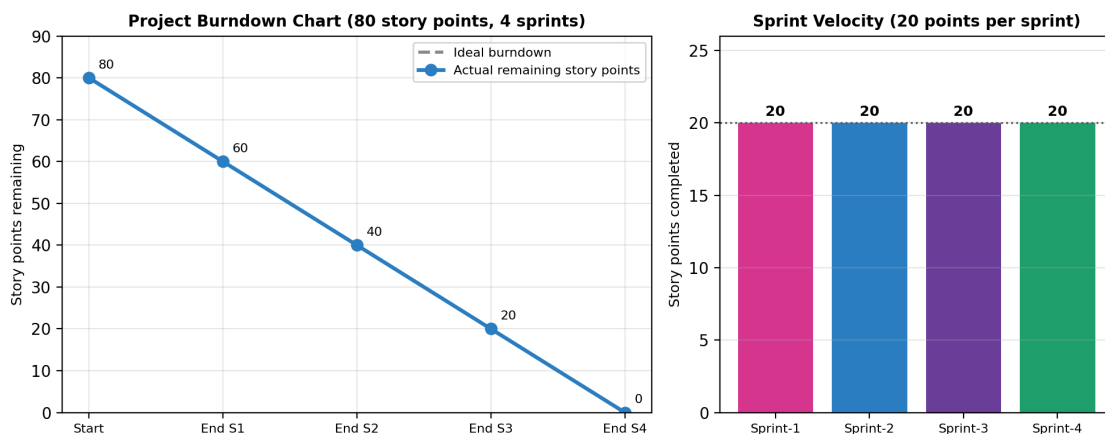


Figure 5.1: Burndown chart (80 points to 0) and sprint velocity

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing covered functional checks of the form, order and flow, performance checks on notification time and multiple requests, and user acceptance testing (UAT).

ID	Scenario	Expected result	Actual result	Result
FT-01	Text input validation (Requested For, Mobile Number, Address)	Valid inputs accepted	Values accepted	Pass
FT-02	Total Amount and Mode of Payment	Amount accepted; UPI / Card selectable	5000 accepted, UPI selected	Pass
FT-03	Order Now creates a request	Order Confirmation and request number	REQ0010004 created, stage Request Approval	Pass
FT-04	Flow / automation check	Flow creates record, emails, asks approval	Flow ran for REQ0010002, 0003, 0004	Pass
PT-01	Response time (request to notification)	Under 3 seconds	About 1 second (02:55:14 to 02:55:15)	Pass
PT-02	Multiple requests processed	No failure or slowdown	3 of 3 requests, 6 of 6 emails	Pass
PT-03	HTML email rendering	Details rendered correctly	'Your request was approved', RITM0010003	Pass

User acceptance testing summary: 10 test cases were defined; 9 passed, 0 failed and 1 (TC-010, negative test with missing mandatory details) was not tested. Two minor defects are open: BG-001 (Medium) – the existing-ID field is still visible when Type of Connection is New, and BG-002 (Low) – notification emails go to sample example.com addresses on the developer instance. No Severity 1 or 2 defects were found.

Section	Total cases	Not tested	Fail	Pass
Catalog Item and Service Portal	2	0	0	2
Form Validation (Catalog UI Policy)	2	1	0	1
Request Submission and Tracking	2	0	0	2
Workflow Automation (Flow Designer)	2	0	0	2
Email Notifications (sys_email)	2	0	0	2
Totals	10	1	0	9

7. RESULTS

7.1 Output Screenshots

The screenshots below were captured from the working solution.

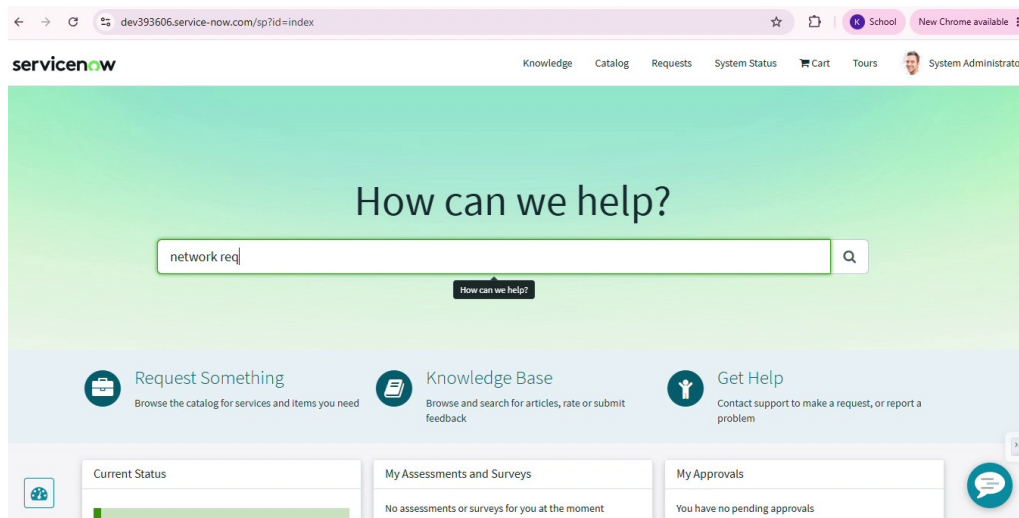


Figure 7.1: Service Portal home page

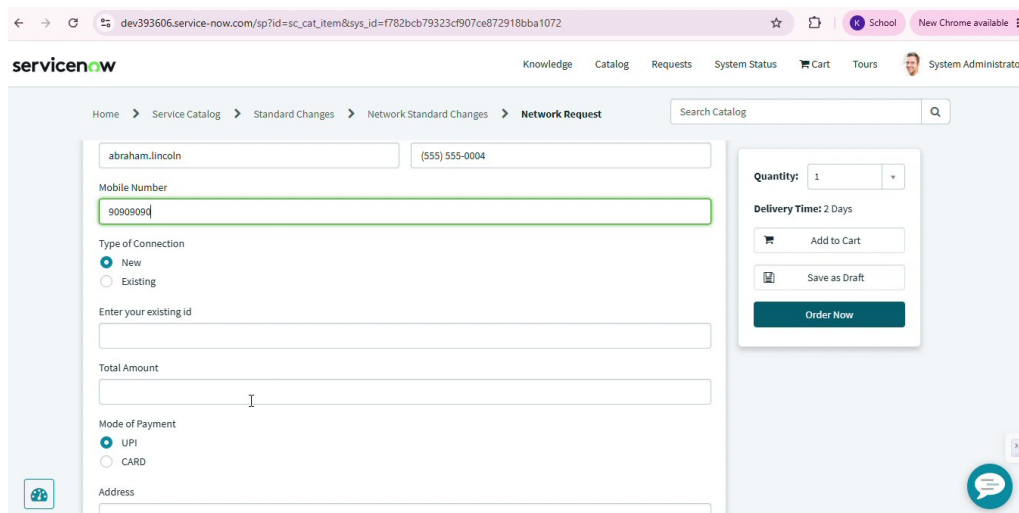


Figure 7.2: Network Request form with Type of Connection, Total Amount, Mode of Payment and Address

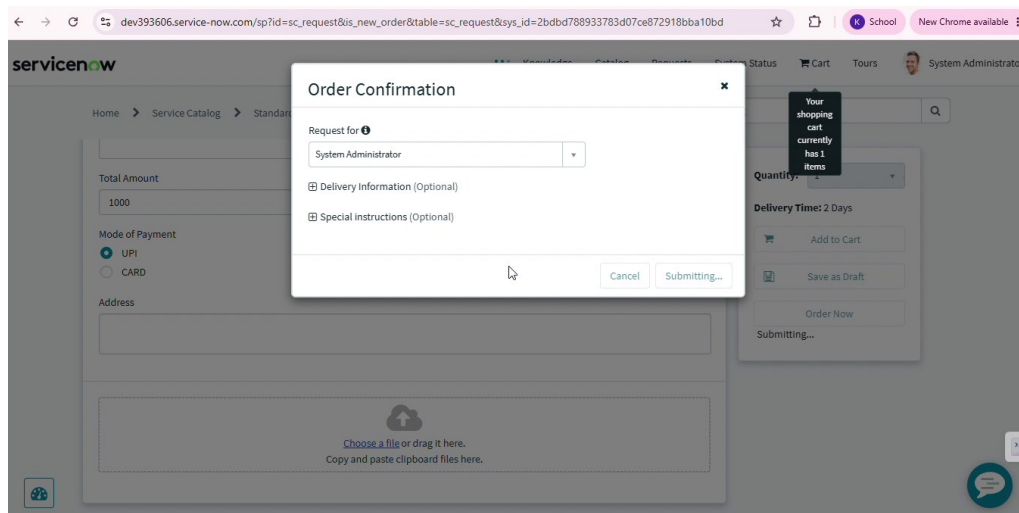


Figure 7.3: Order Confirmation after clicking Order Now

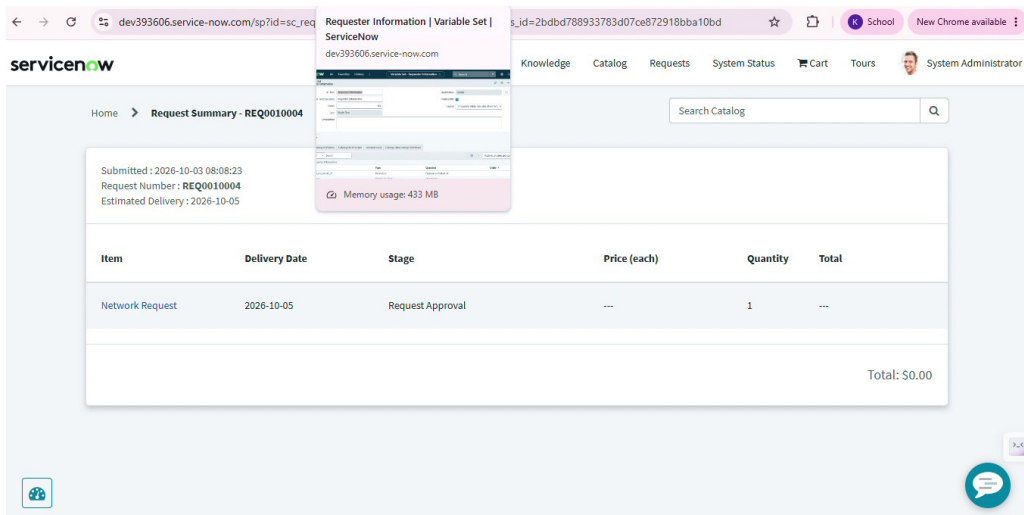


Figure 7.4: Request Summary (REQ0010004) with stage and estimated delivery

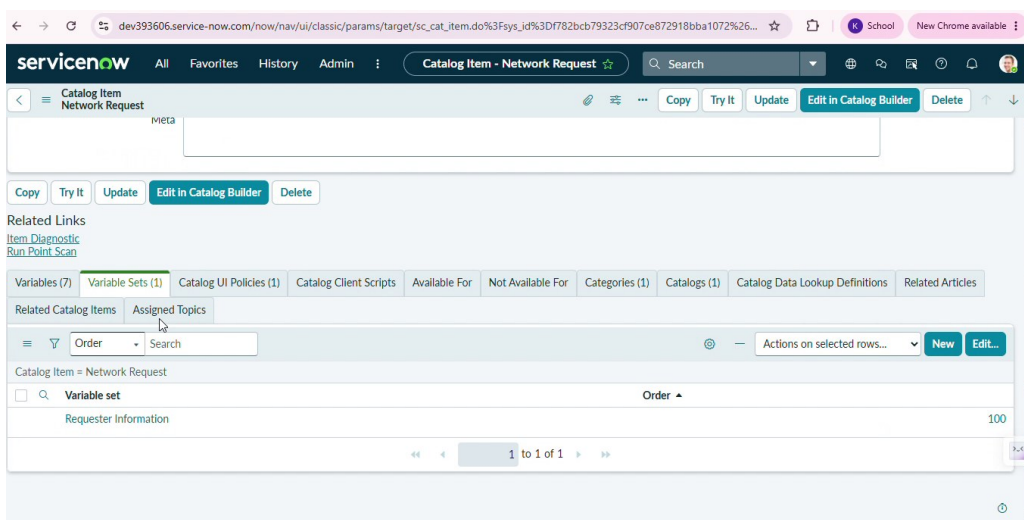


Figure 7.5: Network Request catalog item and Requester Information variable set

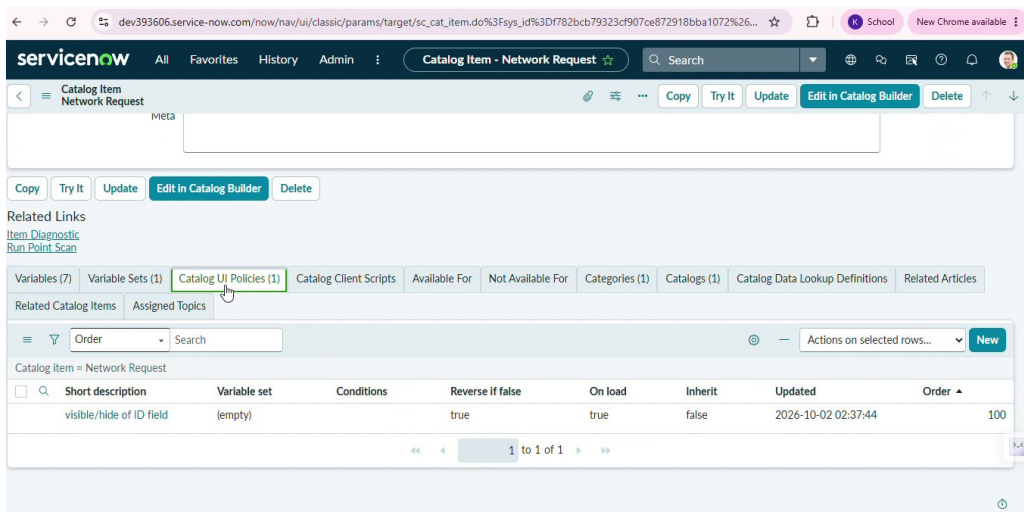


Figure 7.6: Catalog UI Policy 'Hide/Show of ID field'

Figure 7.7 shows the 'Table' configuration page for 'Database Tables' in ServiceNow. The 'Columns' tab is active, displaying a list of columns for the table 'u_database_tables'. The columns are defined as follows:

Column label	Type	Reference	Max length	Default value	Display
Mobile Number	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Mode of Payment	String	(empty)	40		false
Database Number	String	(empty)	40		false
Created by	String	(empty)	40		false
Customer Address	String	(empty)	40		false
Assignment to	Reference	User	32		false

Figure 7.7: Custom table u_database_tables (Database Tables) – columns

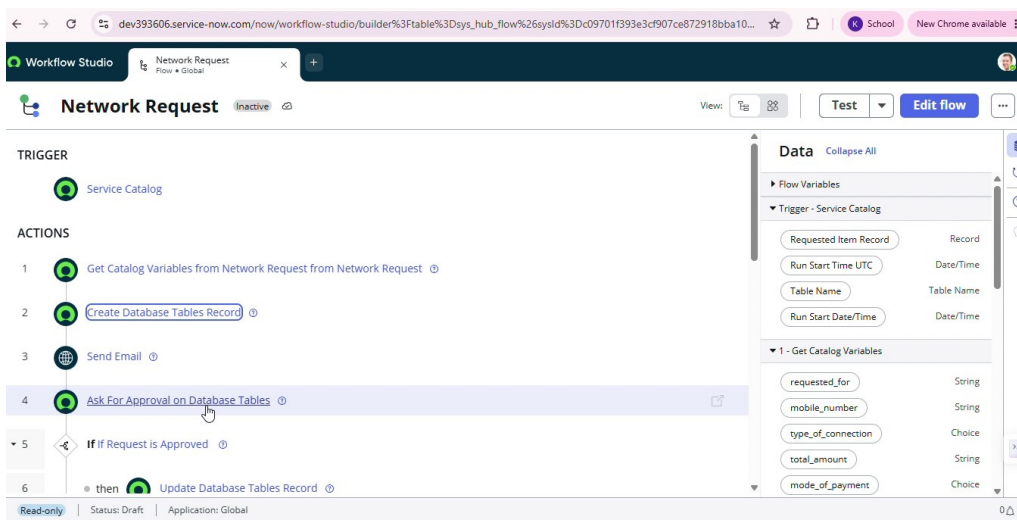


Figure 7.8: Network Request flow in Workflow Studio

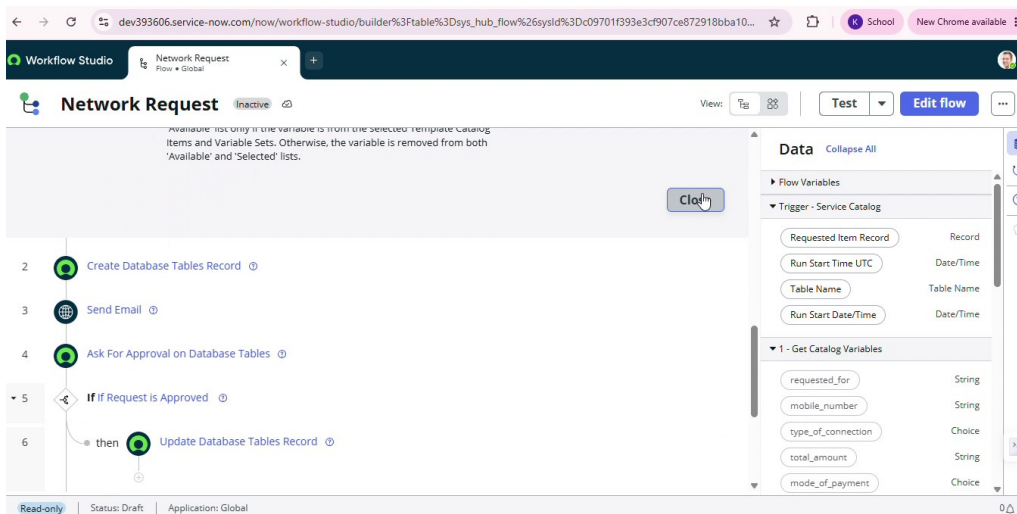


Figure 7.9: Create Database Tables Record – field mapping

Created	Recipients	Subject	Type	Notification type	User ID
2026-10-03 02:49:55	fred.luddy@example.com	Catalog Task SCTASK0010004 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 02:55:15	fred.luddy@example.com	Catalog Task SCTASK0010006 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 08:08:25	fred.luddy@example.com	Catalog Task SCTASK0010008 has been assigned to group Software	send-ready	SMTP	(empty)
2026-10-03 00:00:15	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	send-ready	SMTP	(empty)
about 5 hours ago	admin@example.com	Request REQ0010002 was approved	send-ready	SMTP	(empty)
2026-10-03 02:56:15	admin@example.com	Request REQ0010003 was approved	send-ready	SMTP	(empty)
2026-10-03 08:08:25	admin@example.com	Request REQ0010004 was approved	send-ready	SMTP	(empty)
2026-10-03 02:49:54	admin@example.com	Request REQ0010002 was created	send-ready	SMTP	(empty)
2026-10-03 02:55:14	admin@example.com	Request REQ0010003 was created	send-ready	SMTP	(empty)
2026-10-03 08:08:25	admin@example.com	Request REQ0010004 was created	send-ready	SMTP	(empty)
2026-10-03 01:00:05	alileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	send-ready	SMTP	(empty)
2026-10-03 01:00:06	alileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	send-ready	SMTP	(empty)
2026-10-03 01:00:07	alileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	send-ready	SMTP	(empty)

Figure 7.10: Emails log showing 'created' and 'approved' notifications

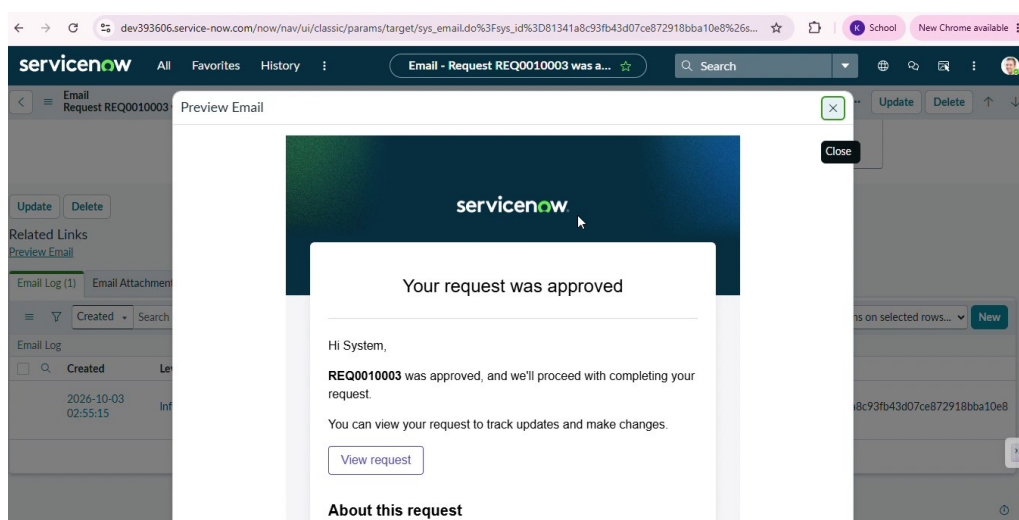


Figure 7.11: Dynamic HTML approval email preview

8. ADVANTAGES & DISADVANTAGES

Advantages

- Standard request form with dynamic validation reduces incomplete and incorrect submissions.
- Records are created automatically with a unique DAT... number, with no manual data entry by IT.
- Approval is routed to the manager automatically and the record is updated on the decision.
- Automated emails and REQ / RITM tracking give employees full visibility, which reduces status enquiries.
- Runs on a cloud platform, so it scales across departments without extra infrastructure.
- Uses only standard enterprise tools; no browser plug-ins are needed.

Disadvantages

- Depends on the ServiceNow platform and its licensing and maintenance.
- Open minor defect: the existing-ID field is still visible for New connections (BG-001).
- Emails currently go to sample addresses on the developer instance (BG-002); real recipients and SMTP must be configured for production.
- The negative test for missing mandatory details (TC-010) is still to be executed.
- Initial configuration needs ServiceNow administration skills.

9. CONCLUSION

The Automated Network Request Management System shows that a network request process can be standardised and automated end to end on ServiceNow. The Network Request catalog item with Catalog UI Policies captures complete information, the Flow Designer flow creates DAT... records, routes approval and updates the record, and automated emails keep the employee informed. Testing showed 9 of 9 executed UAT cases passing, all three test requests processed with both notifications, and a notification time of about one second. With the two minor open defects fixed, the solution is ready to reduce fulfilment time from days to hours and to remove manual IT intervention from the request process.

10. FUTURE SCOPE

- SLA timers and escalation for approvals that stay pending.
- Virtual Agent / chatbot so employees can ask for request status in natural language.
- Dashboards and reports on request volume, approval time and rejection reasons.
- Integration with network devices and IP address management for automatic provisioning.
- Role-based catalog variants for VPN, Wi-Fi and static IP requests with their own approval chains.
- Mobile-friendly request experience and real recipient / SMTP configuration for production.

11. APPENDIX

Item	Details
Source Code (if any)	The solution is configured on the ServiceNow platform (catalog item, variable set, Catalog UI Policy, table u_database_tables and the 'Network Request' flow) and has no separate source code repository.
Dataset Link	Not applicable – data is generated by users through the Network Request form and stored in u_database_tables.
GitHub & Project Demo Link	To be added by the team (the project demo was recorded as a screen recording of the developer instance).